

Problem Set 1

Applied Stats/Quant Methods 1

Due: October 9, 2025

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 9, 2025. No late assignments will be accepted.

Question 1: Education

A school counselor was curious about the average of IQ of the students in her school and took a random sample of 25 students' IQ scores. The following is the data set:

```
1 y <- c(105, 69, 86, 100, 82, 111, 104, 110, 87, 108, 87, 90, 94, 113, 112, 98,
      80, 97, 95, 111, 114, 89, 95, 126, 98)
```

1. Find a 90% confidence interval for the average student IQ in the school.

Solution to question 1.1

```
1 #Question1.1
2 #Sample size
3 n <- length(y)
4 # Sample mean and standard deviation
5 mean_iq <- mean(y)
```

```

6 sd_iq <- sd(y)
7 # Standard error of the mean
8 se <- sd_iq / sqrt(n)
9 #Confidence Intervals
10 # t critical value for 90% CI (two-tailed, df = n-1)
11 t_crit <- qt(0.90, df = n - 1)
12 # Confidence interval
13 lower <- mean_iq - t_crit * se
14 upper <- mean_iq + t_crit * se
15 cat("90% Confidence Interval for mean IQ: [", lower, ", ", upper, "]\n")

```

The 90 percent confidence intervals are [94.99 , 101.89]

- Next, the school counselor was curious whether the average student IQ in her school is higher than the average IQ score (100) among all the schools in the country.

Using the same sample, conduct the appropriate hypothesis test with $\alpha = 0.05$.

Solution to question 1.2

```

1 #Question1.2
2 # One-sided test: is the mean greater than 100?
3 mu0 <- 100 #hypothesized mean
4 test_statistic <- (mean_iq - mu0)/se
5 test_statistic
6 #finding t_crit using alpha = 0.05
7 t_crit <- qt(0.05, df = n - 1, lower.tail = FALSE)
8 t_crit
9 # comparison and decision
10 if(test_statistic > t_crit){
11   decision <- "Reject H0: The school's mean IQ is significantly higher
12     than 100."
13 } else {
14   decision <- "Fail to Reject H0: No evidence that the school's mean IQ
15     is higher than 100."
16 }
17 list(
18   decision = decision
19 )

```

According to my hypothesis test, the test statistic is smaller than the critical value so there is no evidence that the school's mean IQ is higher than 100.

Question 2: Political Economy

Researchers are curious about what affects the amount of money communities spend on addressing homelessness. The following variables constitute our data set about social welfare expenditures in the USA.

State	50 states in US
Y	per capita expenditure on shelters/housing assistance in state
X1	per capita personal income in state
X2	Number of residents per 100,000 that are "financially insecure" in state
X3	Number of people per thousand residing in urban areas in state
Region	1=Northeast, 2= North Central, 3= South, 4=West

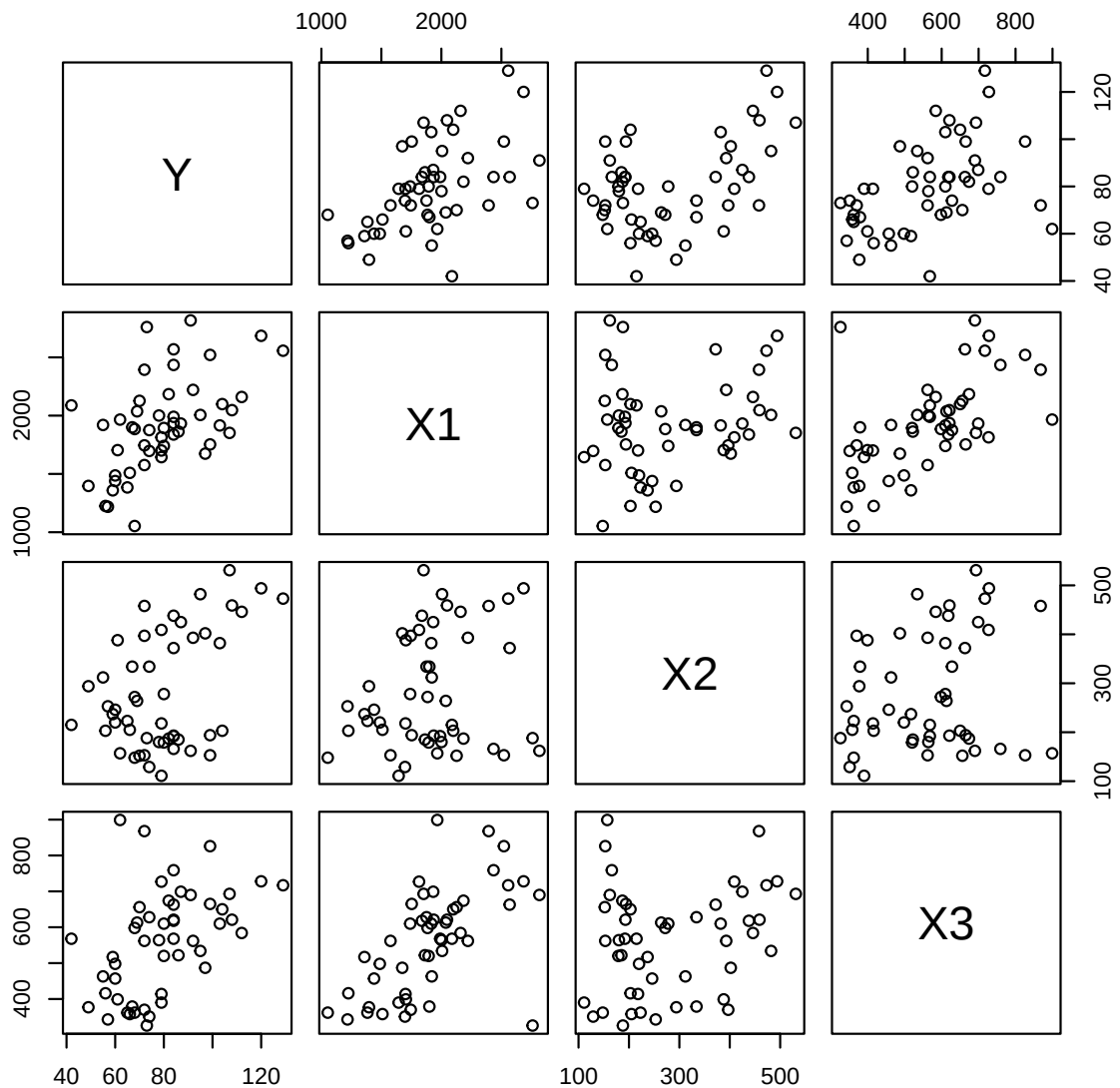
Explore the `expenditure` data set and import data into R.

- Please plot the relationships among Y, X1, X2, and X3? What are the correlations among them (you just need to describe the graph and the relationships among them)?

Solution to question 2.1

```
1 #Question 2.1
2 #read expenditure data
3 expenditure <- read.table("https://raw.githubusercontent.com/ASDS-TCD/
  StatsI_2025/main/datasets/expenditure.txt", header=T)
4 #Explore Data
5 str(expenditure)           # Check structure and variable types
6 summary(expenditure)      # Get descriptive statistics
7 head(expenditure)         # Preview first few rows
8 # Select only the variables of interest
9 exp_data <- expenditure[, c("Y", "X1", "X2", "X3")]
10 # Create the pairs plot
11 pairs(exp_data)
```

Here, I first inspected the data using `str`, `summary` and `head`. "`expdata`" is a subset of the original data set; that includes only the necessary variables. Using the '`pairs`' function I plotted the data. From the graphs below we can observe the following: Most graphs have a positive correlation among them, however, X1 and X2, X2 and X3 don't really have a strong trend. Whereas, Y has a positive trend with all the graphs.



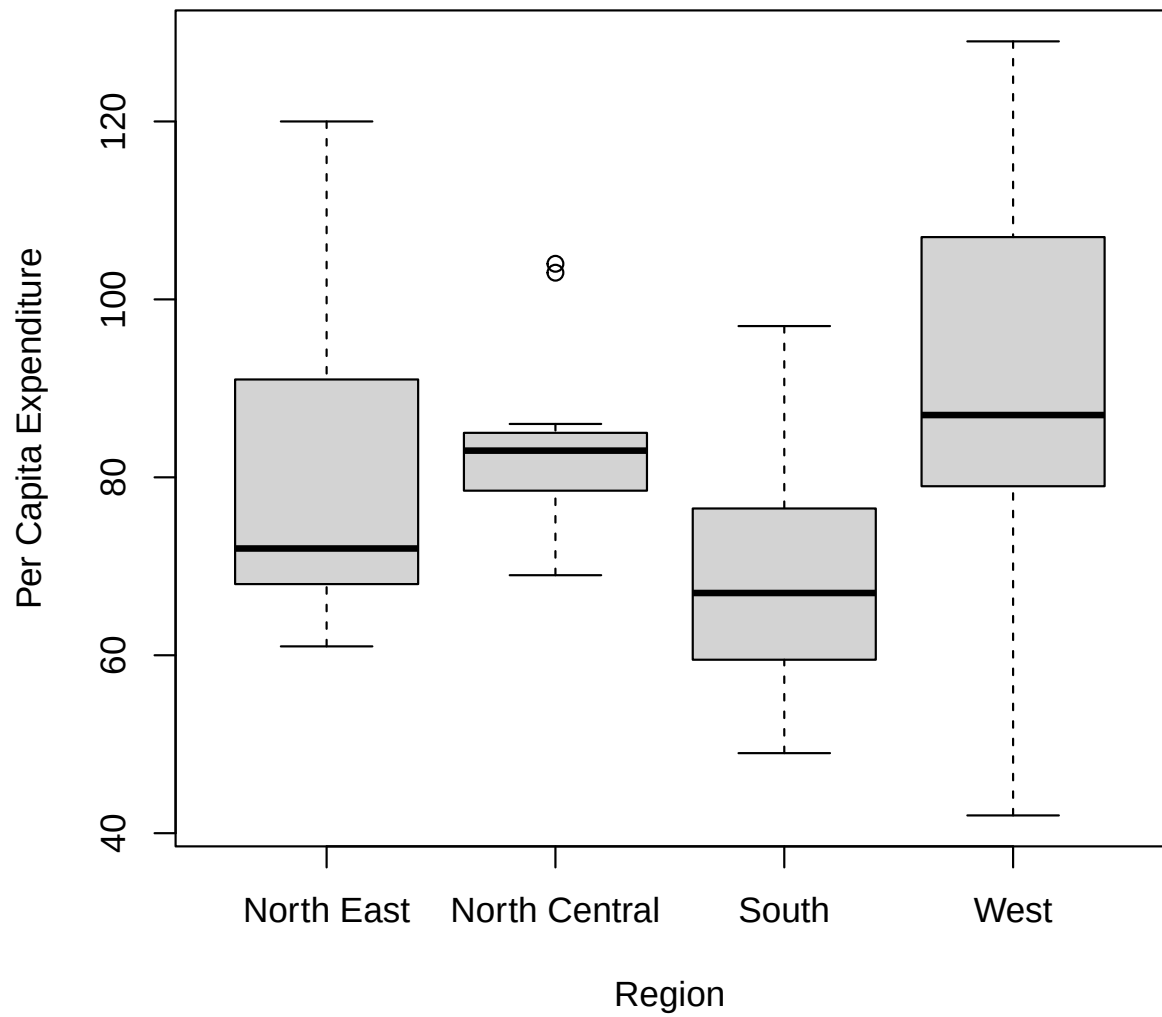
- Please plot the relationship between Y and *Region*? On average, which region has the highest per capita expenditure on housing assistance?

Solution to question 2.2

```

1 #region and plot
2 boxplot(Y ~ Region, data = expenditure,
3         xlab = "Region",
4         ylab = "Per Capita Expenditure",
5         names = c("North East", "North Central", "South", "West"))

```



In this plot, the region which has the highest per capita expenditure on housing assistance on average is West.

- Please plot the relationship between Y and $X1$? Describe this graph and the relationship. Reproduce the above graph including one more variable *Region* and display different regions with different types of symbols and colors.

Solution to question 2.3

```

1 region_colors <- c("pink", "cyan", "darkgreen", "yellow")
2 region_symbols <- c(17, 18, 19, 20)
3
4 #plot
5 expenditure$Region <- factor(expenditure$Region,
6                             levels = c(1,2,3,4),
7                             labels = c("Northeast", "North Central", "South", "
      West"))
8 plot(expenditure$X1, expenditure$Y,
9       main = "Per Capita Expenditure vs Personal Income by Region",
10      xlab = "Per Capita Personal Income (X1)",
11      ylab = "Per Capita Expenditure (Y)",
12      col = region_colors[expenditure$Region],
13      pch = region_symbols[expenditure$Region])
14 #legend
15 legend("topleft",
16       legend = levels(expenditure$Region),
17       col = region_colors,
18       pch = region_symbols,
19       title = "Region")

```

From the graph below we can infer that, there is a noticeable positive linear relationship between the per capita expenditure and the personal income per capita for all regions.

Per Capita Expenditure vs Personal Income by Region

